

Board Game Rating Analysis

Using BoardGameGeek Data

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Code: github.com/lelandweeks/BoardGameGeek-Rating-Analysis

Dataset Overview

~22,000

games (rows)

432

features (48 num + 373 binary)

4

source files merged

FILE

CONTRIBUTES

games.csv

Core game attributes: AvgRating, GameWeight, NumOwned, MfgPlaytime, YearPublished

mechanics.csv

One-hot mechanics (cooperative play, deck-building, dice rolling, etc.)

themes.csv

One-hot themes (fantasy, sci-fi, economic, war, etc.)

subcategories.csv

One-hot subcategories refining game classification

Three Research Questions

Q1

What games are overrated and underrated?

CLASSIFICATION



Q2

Can we predict a game's average rating from its features?

REGRESSION



Q3

Does complexity predict rating within a genre?

SEGMENTED REGRESSION



EDA Highlights

~6.4

MEAN AvgRating

Approximately normal, low skew — a well-behaved target variable

$r = 0.48$

GameWeight $\leftrightarrow$ RATING

Strongest single numeric predictor of average rating

3

FEATURES log1p-TRANSFORMED

MfgPlaytime, NumExpansions, MaxPlayers — severely right-skewed

skew ~74

NumOwned — EXTREME RIGHT SKEW

log1p applied for visualization only; raw ratio used for Q1 label engineering

Feature Engineering

1

Transform skewed features

log1p applied to MfgPlaytime, NumExpansions, MaxPlayers

2

Add 373 binary features

One-hot mechanics, themes & subcategories appended to feature set

3

Scale for linear models

StandardScaler fit on train only — no data leakage; tree models skip scaling

4

Engineer Q1 label

AvgRating / NumOwned → pd.qcut into 3 equal classes

Final feature set: 6 numeric + 373 binary = 379 features

Q1 Results: Classification

0.578

ACCURACY

+25pp above chance (0.33 baseline)

Model: Logistic Regression
(GridSearchCV, best C=0.01)

OVERRATED games are easiest to identify; the middle class is inherently ambiguous by design.

PRECISION / RECALL BY CLASS

Class	Precision	Recall
overrated	0.75	0.66
underrated	0.52	0.67
middle Rated	0.48	0.40

Note: NumOwned was used to engineer the label and was excluded as a feature to prevent data leakage.

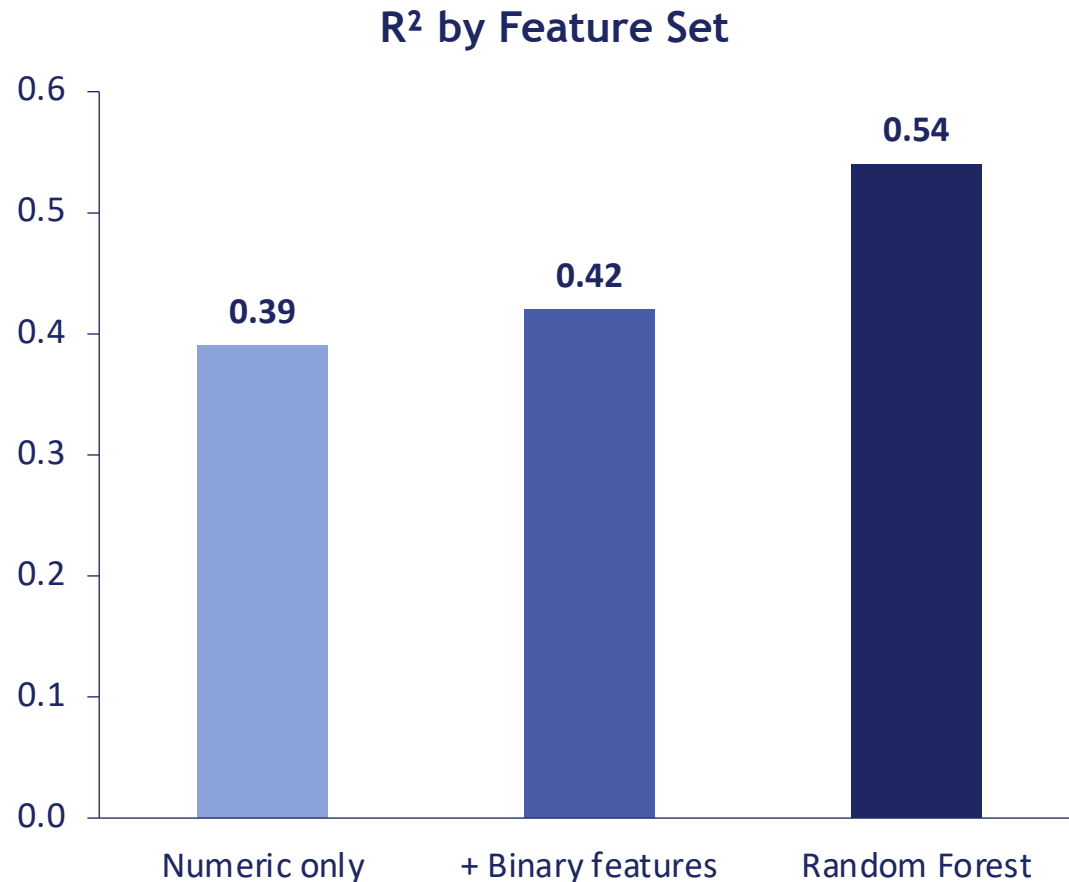
Q2 Results: Model Comparison

PREDICTING AvgRating — RANKED BY R²

Model	R ²	RMSE
Random Forest	0.543	0.649
Lasso ($\alpha=0.001$)	0.425	0.728
Linear Regression	0.424	0.729
Ridge ($\alpha=1.0$)	0.424	0.729
Decision Tree	0.138	0.892

Random Forest substantially outperforms linear models. Decision Tree overfits without depth constraint; Random Forest corrects this via ensemble averaging.

Q2 Key Finding: Feature Importance



+0.12 R² from binary features

Adding 373 one-hot mechanics & themes lifted R² from 0.42 to 0.54

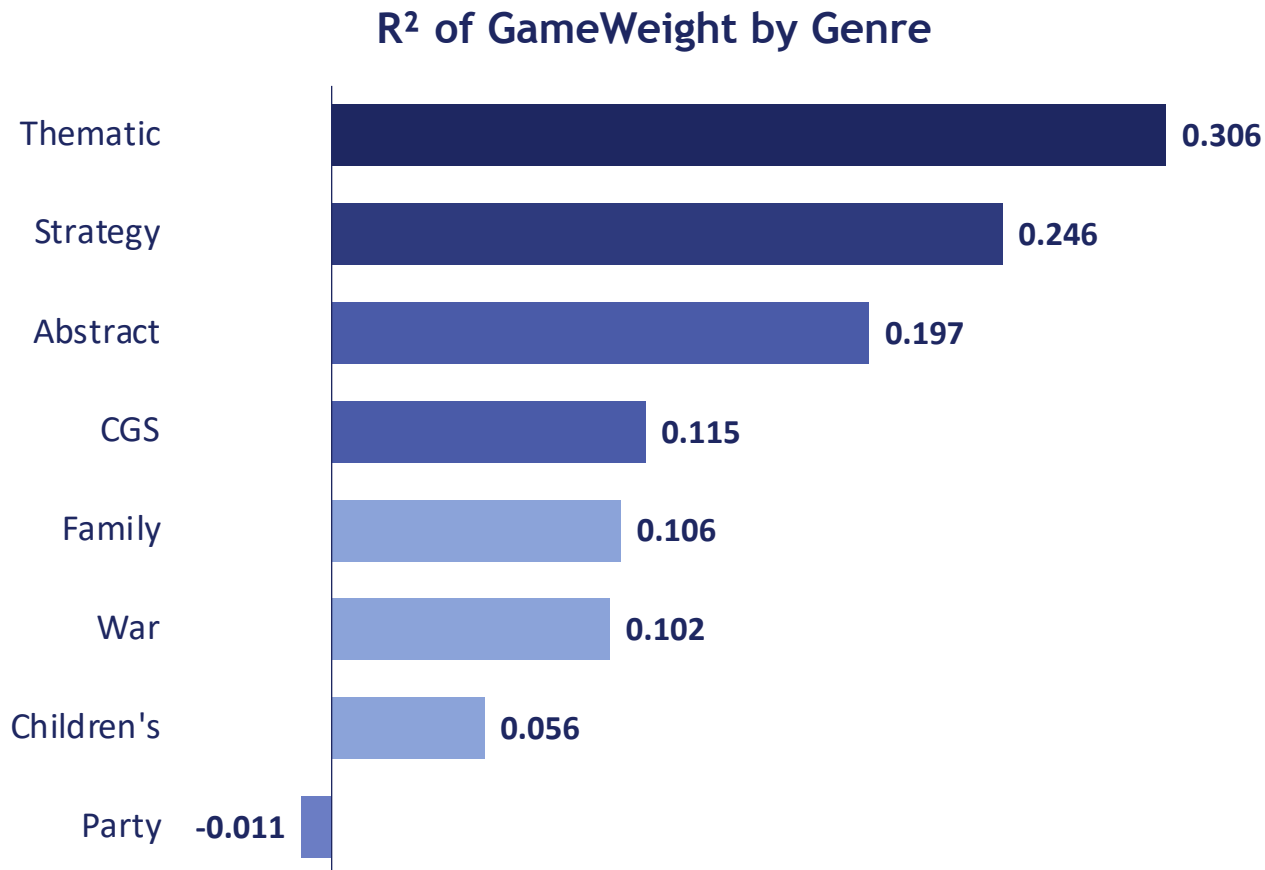
GameWeight: $r = 0.48$

Strongest single numeric correlate of average rating

Design patterns matter

Binary mechanics and themes capture game-design patterns that the numeric features alone simply miss.

Q3 Results: Complexity by Genre



Complexity matters where it's rewarded

Thematic & Strategy games show the clearest complexity–rating link ($R^2 \approx 0.25\text{--}0.31$).

Party games: zero signal

Party games show $R^2 = -0.011$ — complexity has zero predictive value in casual genres.

Discoveries & Limitations

DISCOVERIES

Q1 Ownership gaps reliably surface overrated games (precision 0.75).

Q2 Random Forest best captures rating ($R^2 = 0.54$); binary features drove the biggest improvement.

Q3 Complexity predicts ratings in strategy genres; irrelevant in casual genres.

LIMITATIONS

- Ratings are subjective community opinions, not objective quality scores.
- NumOwned may reflect game age or availability, not just popularity.
- Q1 middle_*rated* class is inherently ambiguous — boundary cases are hard to classify.
- Q2 explains only 54% of variance — subjective taste is hard to model fully.

Key Takeaways

1

Overrated games are identifiable

High ownership with low ratings creates a clear signal (precision 0.75).

2

Random Forest with 432 features best predicts ratings

$R^2 = 0.54$ — binary game mechanics matter.

3

Complexity predicts ratings in strategy genres

Party games show zero signal.